

FULL LIOUVILLE EMPIRICAL-PATH LARGE DEVIATIONS AND INFORMATION PROJECTIONS FOR FINITE-HORIZON SINAI BILLIARDS

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ABSTRACT. The empirical state retains branch labels, transition edges, and actual excursion profiles. Recession is the boundary of a projective family of genuine finite-coordinate LDPs, and diagonal recovery is constructed from admissible excursion sequences rather than assumed as a Gamma limit. Exact terminal cut excursions complete deterministic physical time.

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CONTENTS

1. Round-eight reconstruction: edge-flow compactification and projective epigraph recovery	1
1.1. The exact marked edge-flow state	2
1.2. Finite projections and the projective epigraph theorem	2
1.3. Recession as boundary edge flow	3
1.4. Moment closure without an unbounded reciprocal coordinate	3
1.5. Finite-core approximation and entropy	4
1.6. Spectral versus recession sectors	4
1.7. Exact physical time	4
Scope and verification status	5

1. ROUND-EIGHT RECONSTRUCTION: EDGE-FLOW COMPACTIFICATION AND PROJECTIVE EPIGRAPH RECOVERY

The state now retains the inducing branch, its transition edge, and the actual excursion profile. Recession is obtained from a directed family of genuine finite-coordinate LDPs; no full-sequence Gamma limit is declared in advance.

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1.1. The exact marked edge-flow state. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be the countable first-return graph. An admissible path is a sequence of edges $e_j = (a_j, a_{j+1})$. For an actual excursion point $y_j \in Y_{a_j}$ put

$$K(y_j, a_j) = \frac{1}{r(a_j)} \sum_{\ell < r(a_j)} \delta_{\Theta^\ell \Gamma(y_j)}.$$

For $R_N = \sum_{j < N} r(a_j)$ define

$$M_N = R_N^{-1} \sum_{j < N} r(a_j) \delta_{(a_j, a_{j+1})}, \quad \Xi_N = R_N^{-1} \sum_{j < N} r(a_j) \delta_{(a_j, a_{j+1}, K(y_j, a_j))}.$$

The mark contains the actual profile, not its branch conditional expectation.

Choose finite edge sets $\mathcal{E}_m \uparrow \mathcal{E}$ and finitely many continuous profile tests $\phi_1, \dots, \phi_m$. The projection $\pi_m \Xi$ records the masses of edges in $\mathcal{E}_m$, their incoming and outgoing marginals, the first m profile moments, and the escaped clock

$$z_m = 1 - \sum_{e \in \mathcal{E}_m} \Xi(e, \mathcal{Z}).$$

Lemma 1.1 (Rate data are functions of the state). *The edge marginals of Ξ determine admissibility, stationarity, the length-weighted entropy flow, and the induced potential average. Distinct branches with the same return length and profile remain distinct coordinates.*

Proof. Incoming and outgoing marginals are read from the first two components of the mark. Stationarity is equality of these marginals. Conditional transition probabilities are the Radon–Nikodym ratios of the edge flow with respect to its incoming marginal, so their entropy is determined by Ξ . The branch potential and length are edge labels and are integrated directly. No injectivity of $a \mapsto (r(a)^{-1}, K_a)$ is used. $\square$

1.2. Finite projections and the projective epigraph theorem. Let I_m be the good finite-dimensional rate obtained by the exponential change of measure on $\pi_m \Xi_N$. The source class contains all edge indicators in $\mathcal{E}_m$ and all profile tests $\phi_1, \dots, \phi_m$. Consistency gives

$$I_m(z) = \inf \{I_{m+1}(z') : \pi_m z' = z\}.$$

Theorem 1.2 (Projective epigraph LDP). *Assume exponential tightness of the return-length moment and the finite-source coefficient theorem of A2. Then Ξ_N satisfies a full good LDP on the projective state space with rate*

$$I(\Xi) = \sup_m I_m(\pi_m \Xi).$$

Every finite-rate state has a diagonal sequence of actual admissible paths whose costs converge to $I(\Xi)$.

Proof. The upper bound follows from exponential tightness and the finite-projection upper bounds. For the lower bound, choose $m_j \uparrow \infty$ so that $I_{m_j}(\pi_{m_j} \Xi) \uparrow I(\Xi)$. The finite-dimensional lower theorem provides an actual

admissible path block in a 2^{-j} neighborhood of $\pi_{m_j}\Xi$ with cost at most $I_{m_j} + 2^{-j}$. Concatenate these blocks as successive excursion sequences. If the terminal edge of one block does not meet the initial edge of the next, insert a finite connecting excursion path supplied by transitivity of the return graph. Every intermediate base return is a legal edge-flow transition rather than a forbidden connector inside one first-return word. Choose block multiplicities increasing so fast that all connecting paths and earlier blocks have vanishing clock fraction. The diagonal path converges in every projection and has the desired cost. This proves the recovery statement rather than assuming it. $\square$

1.3. Recession as boundary edge flow. The boundary mass z_m is monotone in m and converges to $z_\infty \in [0, 1]$. A boundary profile is a projective limit of the moments carried by edges escaping every finite set. Define

$$I_{\text{rec}}(z) = \sup_m I_m^{\text{boundary}}(\pi_m z).$$

Theorem 1.3 (Actual-excursion reversion recovery). *The reversion functional is lower semicontinuous, convex under concatenation of excursion blocks, and has the dual formula*

$$I_{\text{rec}}(z) = \sup_{F \in C_{\text{cyl}}} \{z(F) - q_\infty(F) + q_\infty(0)\}.$$

For every finite-rate z there are actual long first-return excursions whose finite profile projections approach z and whose logarithmic costs approach $I_{\text{rec}}(z)$.

Proof. Each finite boundary rate is a closed convex conjugate and the directed supremum is lower semicontinuous. The recovery part of Theorem 1.2, applied to states with clock mass outside $\mathcal{E}_m$, produces actual excursions at each finite stage. A diagonal selection makes the return length tend to infinity while matching every profile projection. Convex combinations are realized by consecutive long excursions separated by genuine base returns; no claim is made that their concatenation is one first-return word. The cylinder dual determines the projective supremum by separation. $\square$

1.4. Moment closure without an unbounded reciprocal coordinate. For every empirical state,

$$\Xi_N(1) = 1.$$

The recurrent clock in a finite set and its complement add exactly to one. The boundary coordinate is therefore a continuous ledger coordinate, not the weak limit of the unbounded function x^{-1} .

Lemma 1.4 (Exact recurrent-defect normalization). *Every projective limit satisfies*

$$m_{\text{rec}}(1) + z_\infty = 1.$$

Conversely every stationary edge flow with this identity is approximated by normalized empirical edge flows.

Proof. For each m the identity $m_N(\mathcal{E}_m) + z_{N,m} = 1$ is exact and involves bounded coordinates. Pass first to $N \rightarrow \infty$ and then $m \rightarrow \infty$. For the converse, truncate the flow, decompose the finite circulation into cycles, and append boundary excursion blocks supplied by Theorem 1.3. Normalize by total clock. $\square$

1.5. Finite-core approximation and entropy.

Theorem 1.5 (Stationary flow approximation). *Every finite-rate stationary countable edge flow is approximated by stationary flows on finite strongly connected subgraphs, with convergence of potential, length moment, and entropy. Periodic components retain their period.*

Proof. Truncate to edges carrying all but ε of the length, potential, and information moments. The imbalance at the retained vertices has total mass $O(\varepsilon)$. Decompose the original stationary flow into directed cycles and keep finitely many cycles that repair this imbalance. Their union is a finite graph; apply the construction separately to each cyclic class instead of adding a period-breaking orbit. The entropy error is controlled by the truncated information moment and tends to zero. $\square$

1.6. Spectral versus recession sectors.

Theorem 1.6 (SPR/recession dichotomy). *A strongly positive recurrent component contributes through its isolated renewal pole and the recurrent edge-flow rate. If the pole is lost at the critical pressure, its normalized mass escapes every finite edge set and is represented by I_{rec} . No component is counted in both sectors.*

Proof. On a finite core the renewal resolvent is compact and has an isolated Perron pole. Uniform return-length integrability keeps this pole isolated in the countable limit, which is the SPR case. If uniform integrability fails, any normalized approximate eigenvector has vanishing mass on every finite edge set; its finite projections converge to a boundary state. The projective epigraph rate of that state is exactly the recession cost. The alternatives are mutually exclusive by tightness. $\square$

1.7. Exact physical time. For a deterministic time T , keep the terminal cut excursion $(a_*, u_*, K_*^{\text{cut}})$, where $u_* \in [0, 1]$ is the used fraction of its roof. This coordinate is part of the state rather than discarded.

Theorem 1.7 (Deterministic-time path LDP). *The edge-flow LDP contracts to the full empirical path at every deterministic physical time. The terminal cut coordinate gives exact time T and its cost is the corresponding fractional excursion cost. When the terminal roof is $o(T)$ it is exponentially equivalent to zero; when it is of order T it is already represented by the recession sector.*

Proof. Stop at the first return whose accumulated roof exceeds T and retain the used fraction of that excursion. This makes the clock identity exact.

Exponential tightness separates the event $r(a_*) \leq \delta T$, whose profile error is at most δ , from $r(a_*) > \delta T$, which is a boundary excursion and is governed by Theorem 1.3. Let $\delta \downarrow 0$ after the LDP bounds. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-eight dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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